C6.3 Interpreting and interacting with earth systems

Summary

As our understanding of the structure of materials and chemical processes has improved we are increasing our ability to interpret and understand chemical and earth systems. Understanding how we interact with them is very important to our survival as a species. This section starts with the history of the atmosphere and moves on to how human activity could be affecting its composition.

Underlying knowledge and understanding

Learners should have some understanding of the composition of the Earth, the structure of the Earth, the rock cycle, the carbon cycle, the composition of the atmosphere and the impact of human activity on the climate.

Common misconceptions

Learners think that the atmosphere is large and that small increases of carbon dioxide or a few degrees of temperature change do not make a difference to the climate. They may consider that global warming is caused by the ozone hole and that human activities alone cause the greenhouse effect.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM6.3i	extract and interpret information from charts, graphs and tables	M2c, M4a
CM6.3ii	use orders of magnitude to evaluate the significance of data	M2h

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	**To include**	**Maths**	**Working scientifically**	
C6.3a interpret evidence for how it is thought the atmosphere was originally formed	knowledge of how the composition of the atmosphere has changed over time	M2c, M4a, M2h	WS1.3e	
C6.3b describe how it is thought an oxygen-rich atmosphere developed over time		M2h	WS1.1a	
C6.3c describe the greenhouse effect in terms of the interaction of radiation with matter within the atmosphere				
C6.3d evaluate the evidence for additional anthropogenic (human activity) causes of climate change and describe the uncertainties in the evidence base	the correlation between change in atmospheric carbon dioxide concentration and the consumption of fossil fuels	M2c, M4a, M2h		
C6.3e describe the potential effects of increased levels of carbon dioxide and methane on the Earth's climate and how these effects may be mitigated	consideration of scale, risk and environmental implications	M2c, M4a, M2h	WS1.1f, WS1.1h	
C6.3f describe the major sources of carbon monoxide, sulfur dioxide, oxides of nitrogen and particulates in the atmosphere and explain the problems caused by increased amounts of these substances			WS1.4a	
C6.3g describe the principal methods for increasing the availability of potable water in terms of the separation techniques used	ease of treatment of waste, ground and salt water			